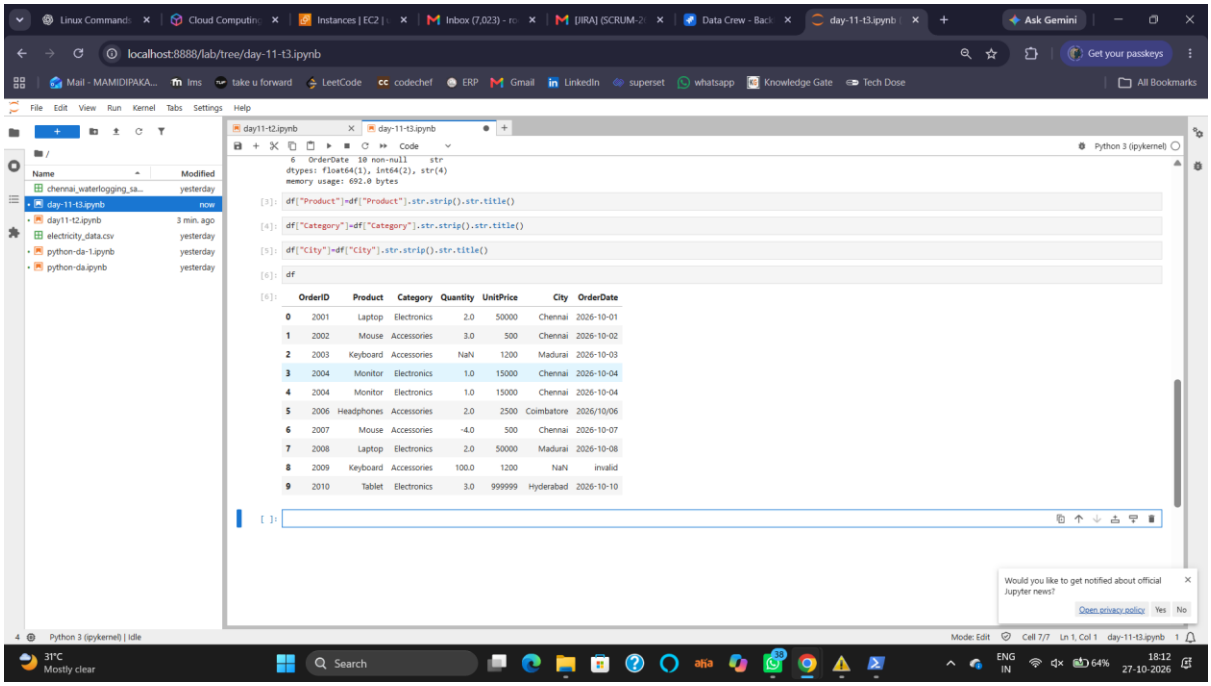
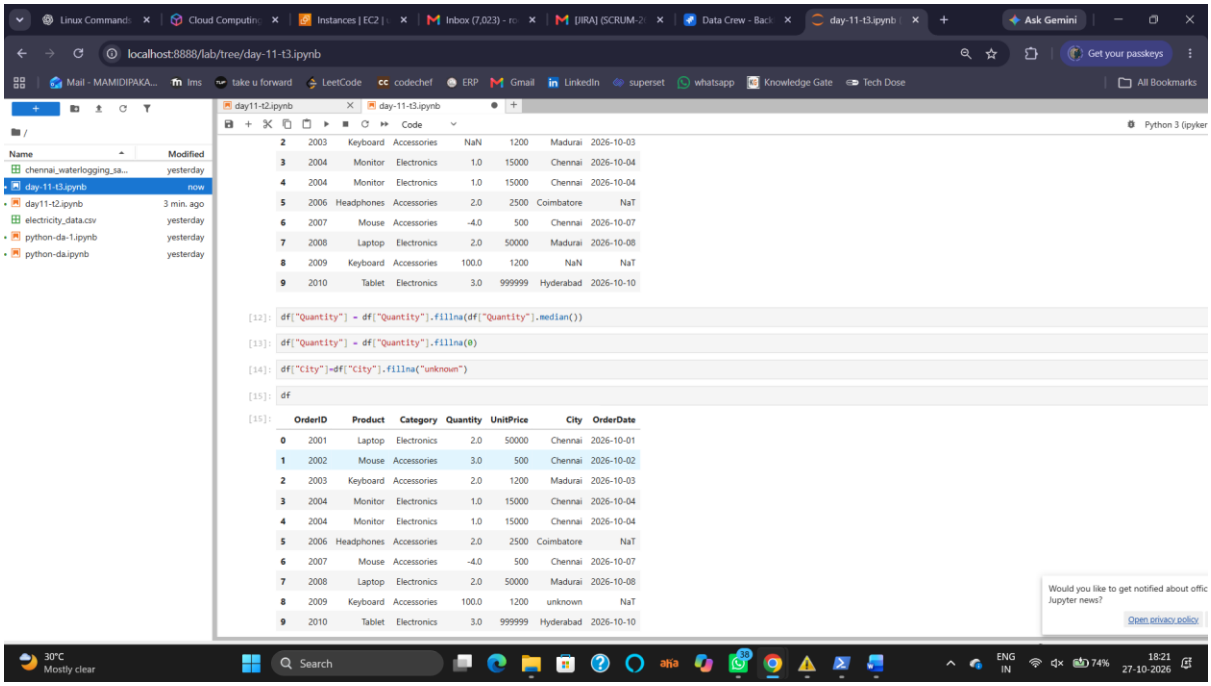


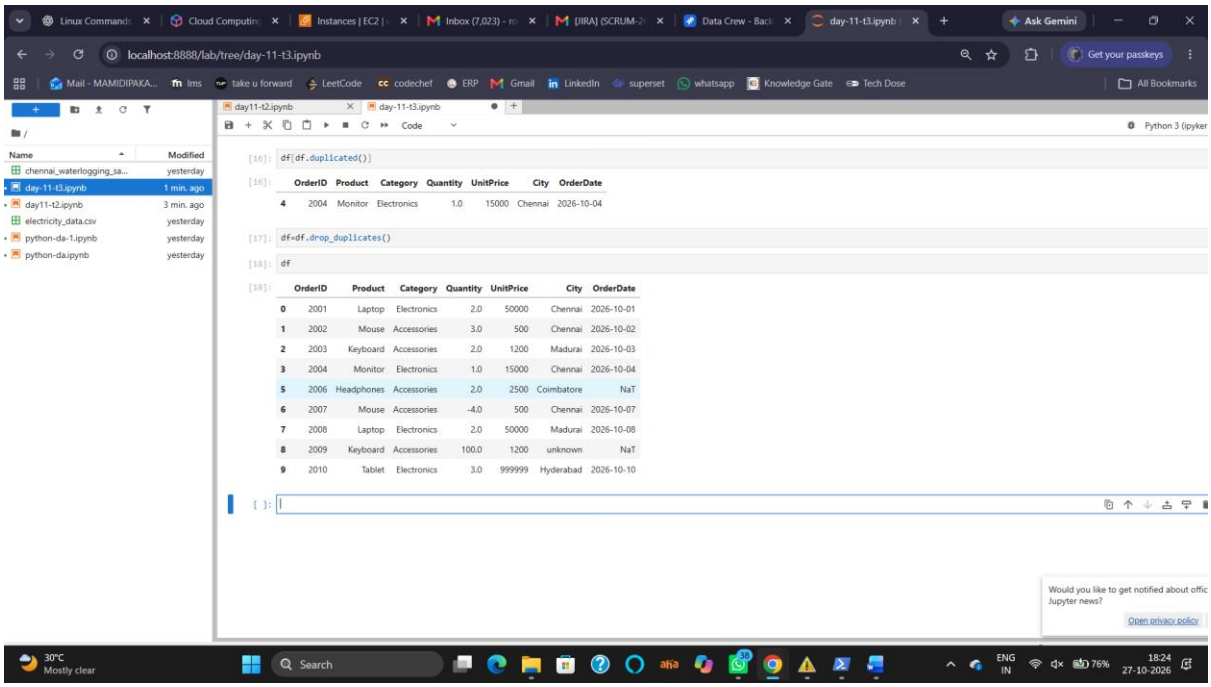
Data formatting

Changing the format of data into a consistent and appropriate format improves the accuracy and efficiency of business operations. For example, converting order dates into a proper datetime format allows the business to analyze daily, monthly, and yearly sales trends, while converting quantity and price columns into numeric formats enables accurate calculations of revenue, costs, and profits. Standardizing text formats, such as product names and city names, prevents the same product or location from being treated as different entries due to capitalization or extra spaces. These changes make filtering, sorting, grouping, and reporting more reliable, reduce calculation errors, and help management identify sales trends and make better business decisions.



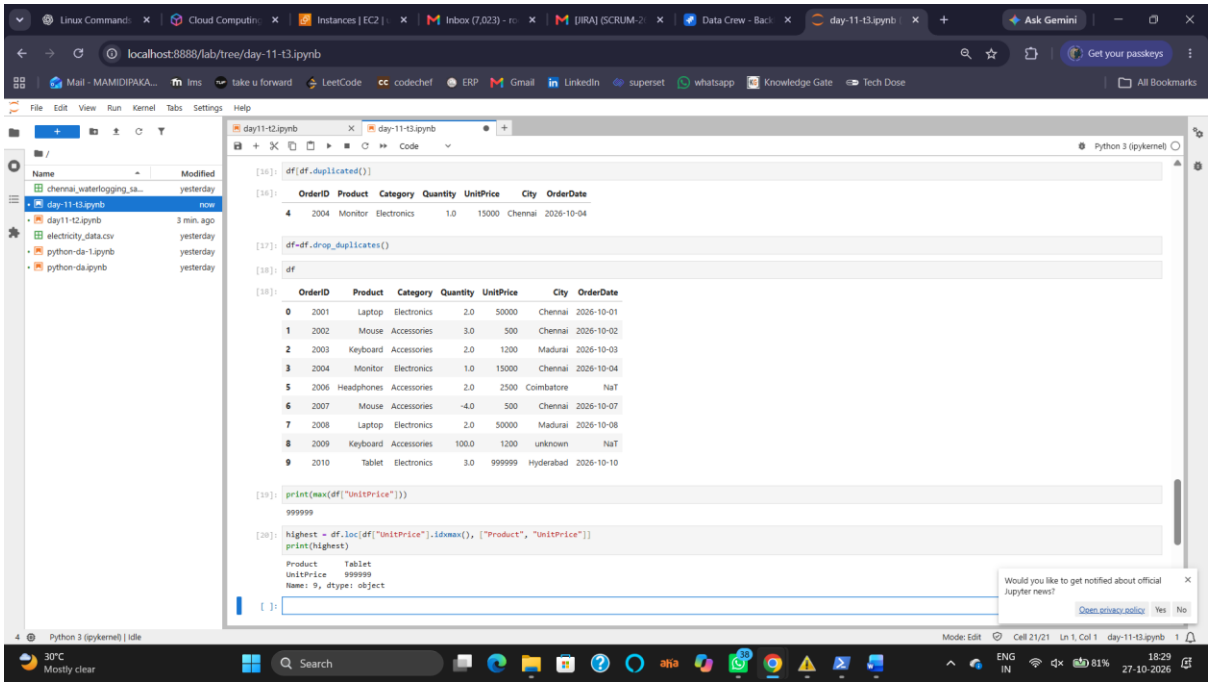
Filling missing values

Filling missing values improves the completeness and reliability of business data, enabling the company to perform accurate sales analysis and make better decisions. For example, filling missing quantities helps calculate total sales correctly, while filling missing product prices supports revenue estimation and financial reporting. Replacing missing city names with "Unknown" helps preserve records for further analysis, and recovering missing order dates improves sales trend analysis and order tracking. Properly handling missing values also reduces errors in reports, improves inventory planning, and helps management identify customer demand more effectively. However, values should be filled using appropriate methods or verified information to avoid introducing incorrect data that could lead to poor business decisions.



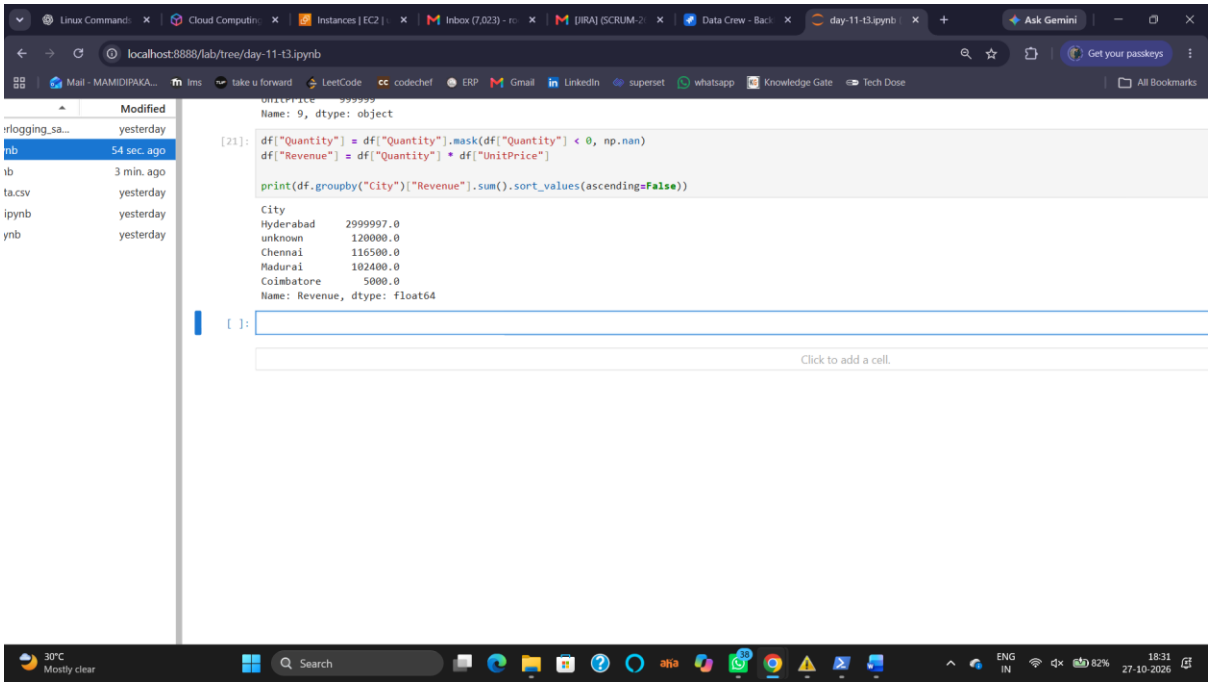
Removing duplicate records

Removing duplicate records improves the accuracy and reliability of business data by ensuring that the same order or transaction is not counted multiple times. Duplicate records can inflate total sales revenue, distort product demand, and lead to incorrect inventory planning and financial reports. By identifying and removing duplicates, the business can calculate sales more accurately, maintain consistent customer and order information, reduce unnecessary operational costs, and make better data-driven decisions. This also helps management gain a more realistic understanding of business performance and avoid mistakes in forecasting and resource allocation. However, duplicates should be removed carefully by verifying unique identifiers and business rules to avoid accidentally deleting legitimate transactions.



Finding highest priced product

Identifying the highest-priced product helps the business understand its premium product segment and make better pricing and sales decisions. It allows management to evaluate high-value products, estimate their contribution to potential revenue, and develop suitable marketing strategies to target customers willing to spend more. This information also supports inventory planning, profit-margin analysis, and decisions about discounts and promotional offers. By comparing the highest-priced product with other products, the business can identify opportunities to improve profitability and position its premium products more effectively in the market. However, the highest price alone does not indicate the highest profit, as factors such as product cost, sales volume, and customer demand must also be considered.



Grouping by city

Grouping sales data by city helps the business understand sales performance and customer demand across different geographical locations. By analyzing total sales, order quantities, and product demand for each city, management can identify high-performing and low-performing markets, allocate inventory more efficiently, and plan targeted marketing campaigns. It also helps the business identify locations with growth opportunities, optimize product distribution, and improve delivery and resource planning. For example, if Chennai generates higher sales than other cities, the business may consider maintaining more inventory there, while cities with lower sales may benefit from targeted promotions. Overall, grouping by city supports location-based decision-making, improves operational efficiency, and helps the business identify opportunities to increase revenue.